

# Experiment 03: Input & Output Manual

## Design a Homepage using React.js

### 🎓 Student & Laboratory Credentials

- **Student Name:** Asmit Jogdand
- **Roll Number:** 25CE1051
- **Class / Division:** B.Tech Computer Engineering (Sem VI)
- **Academic Year:** 2026 – 2027
- **Course:** Skill Based Lab - Advanced Web Technology (SBL-AWT)
- **Institution:** Ramrao Adik Institute of Technology (RAIT), D Y Patil Deemed to be University, Nerul, Navi Mumbai
- **Problem Statement Domain:** HealthPulse — Digital Healthcare & Clinical Management Portal
- **Experiment Title:** Design a Homepage using React.js

## 1. Experiment Aim

To design, implement, and test a component-driven, responsive clinical healthcare portal homepage using **React.js** (React 18), leveraging functional components, props, state management ( `useState` ), JSX, and dynamic category filtering.

## 2. 📁 Input Specification

### 2.1 File System Input

| File Name    | Format                      | Role & Implementation Responsibility                                                                                                          |
|--------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| index.html   | React 18 (Babel Standalone) | Zero-setup Single Page Application root containing React component hierarchy, medical datasets, dynamic state hooks, and inline CSS3 styling. |
| package.json | JSON                        | Project metadata, build scripts, and dependencies configuration.                                                                              |

### 2.2 Execution Command Input

```
# Navigate to the experiment directory
cd /home/ubuntu/SBL-EXPT/Asmit_25CE1051/Expt-03-React-Homepage

# Option A: Open directly in Google Chrome / Chromium browser
chromium-browser index.html

# Option B: Run via lightweight local web server
npx serve .
# Or via Python standard library:
python3 -m http.server 3003
```

## 2.3 Component Architecture & Props Input Structure

| COMPONENT NAME    | STATE HOOKS USED         | PROPS RECEIVED        | FUNCTIONALITY            |
|-------------------|--------------------------|-----------------------|--------------------------|
| App               | appointmentModalVisible  | None (Root Component) | Manages booking triggers |
| StudentHeader     | None (Pure Functional)   | None                  | Renders academic banner  |
| Navbar            | mobileMenuOpen           | onBookClick: function | Brand nav & sticky CTA   |
| HeroSection       | None (Pure Functional)   | onBookClick: function | Headline & stats anchor  |
| StatsBar          | None (Pure Functional)   | stats: Array          | Clinical impact metrics  |
| ServicesSection   | selectedCategory (State) | None                  | Department category tabs |
| ServiceCard       | None (Pure Functional)   | service: Object       | Individual specialty UI  |
| SpecialistSection | None (Pure Functional)   | doctors: Array        | Physician credentials    |
| EmergencyBanner   | None (Pure Functional)   | None                  | 24/7 hotline display     |
| Footer            | None (Pure Functional)   | None                  | RAIT credentials, hours  |

## 2.4 Medical Data Input Vectors

Services Dataset ( `servicesData` ):

```
[
  { id: 1, name: "Cardiology & CathLab", category: "Critical Care", icon: "♥", doctors: 14, waitTime: "10 mins" },
  { id: 2, name: "Neurology & Brain Spine", category: "Critical Care", icon: "a", doctors: 8, waitTime: "15 mins" },
  { id: 3, name: "Orthopedics & Joint Center", category: "Outpatient", icon: "a", doctors: 12, waitTime: "20 mins" },
  { id: 4, name: "Pediatrics & Child Care", category: "Outpatient", icon: "a", doctors: 10, waitTime: "10 mins" },
  { id: 5, name: "Automated Clinical Pathology", category: "Diagnostics", icon: "a", doctors: 6, waitTime: "5 mins" },
  { id: 6, name: "3T MRI & High-Speed CT", category: "Diagnostics", icon: "a", doctors: 5, waitTime: "15 mins" }
]
```

User Interaction Input Triggers:

1. Category Tab Selection: User clicks `"All"`, `"Critical Care"`, `"Outpatient"`, or `"Diagnostics"`.
2. Booking Trigger: User clicks `"Book Appointment"` button on the Hero banner or Navbar.
3. Patient Booking Input: User inputs Patient Name `Asmit Jogdand` and selects Department `Cardiology`.

### 3. Output Specification

#### 3.1 Graphical User Interface Output Representation

```
+-----+
| 🏥 RAIT - Department of Computer Engineering | SBL-AWT Sem VI | Student: Asmit Jogdand (25CE1051)|
+-----+
| [ ❤️ HealthPulse Portal]   Services   Specialists   Emergency   About       [ 📅 Book Appt ] |
+-----+
|
|           📌 ADVANCED CLINICAL HEALTHCARE & SURGICAL EXCELLENCE
|           Empowering Patients with World-Class Healthcare & Digital Telehealth
|
|           [ 45,000+ Patients ]   [ 120+ Specialists ]   [ 24/7 Trauma ]   [ 99.8% Recovery ]
|
+-- Clinical Specialties & Departments -----+
| Filter Categories: [ (•) All ]   [ Critical Care ]   [ Outpatient ]   [ Diagnostics ]
|
| +-----+ +-----+ +-----+
| | ❤️ Cardiology | | 📌 Neurology | | 📌 Orthopedics | |
| | 14 Specialists | | 8 Specialists | | 12 Specialists | |
| | Wait: 10 mins | | Wait: 15 mins | | Wait: 20 mins | |
| +-----+ +-----+ +-----+
|
+-- Board-Certified Clinical Specialists -----+
| 📌 Dr. Amarsinh V. Vidhate   Chief Medical Director & Interventional Cardiologist | |
| 📌 Dr. Radhika Sen           Head of Neurosciences & Stroke Management           | |
+-----+
|
| 📌 24/7 CRITICAL TRAUMA & AMBULANCE HOTLINE: 1800-432-5873 (TOLL-FREE RAPID RESPONSE)
|
+-----+
```

#### 3.2 Dynamic State Category Filtering Output

When the user interacts with the Category Filter Tabs, React updates `selectedCategory` state and re-renders only the relevant service cards via the Virtual DOM:

```
+-----+
| ACTIVE TAB CLICKED | REACT STATE UPDATED | CARDS RENDERED TO UI |
+-----+
| "All" (Default)    | selectedCategory = "All" | All 6 Clinical Department Cards |
| "Critical Care"    | selectedCategory = "Critical Care" | Cardiology & CathLab, Neurology |
| "Outpatient"       | selectedCategory = "Outpatient" | Orthopedics, Pediatrics |
| "Diagnostics"      | selectedCategory = "Diagnostics" | Clinical Pathology, 3T MRI & CT |
+-----+
```

#### 3.3 Appointment Booking Modal Dialog Output

When clicking **"Book Appointment"**, the interactive booking popup renders with auto-generated token:

```
+-----+
|           📌 HEALTHPULSE CLINICAL APPOINTMENT DISPATCH           |
+-----+
| Patient Name:      Asmit Jogdand |
| Preferred Department: Cardiology & CathLab |
| Consulting Physician: Dr. Amarsinh V. Vidhate |
| Generated Queue Token: HP-APPT-88219 |
| Status:            SLOT RESERVED (Confirmation SMS & Email Queued) |
|
| [ OK / Print Token ]
+-----+
```

### 3.4 Test Case Execution Results

| Test Scenario         | User Action Injected      | React Execution & Virtual DOM Behavior                                   | Output Result Observed                                          | Status |
|-----------------------|---------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------------|--------|
| Initial Page Mount    | Page loads in browser     | <code>ReactDOM.createRoot</code> renders <code>&lt;App /&gt;</code> tree | Complete Homepage displays with all 6 specialties               | PASS   |
| Filter: Critical Care | Click "Critical Care" tab | <code>setSelectedCategory("Critical Care")</code>                        | Non-critical departments unmount; Cardiology & Neurology remain | PASS   |
| Filter: Diagnostics   | Click "Diagnostics" tab   | <code>setSelectedCategory("Diagnostics")</code>                          | Only Pathology and MRI cards are displayed                      | PASS   |
| Reset Filter          | Click "All" tab           | <code>setSelectedCategory("All")</code>                                  | All 6 cards re-render cleanly without page reload               | PASS   |
| Book Appointment      | Click Hero CTA button     | Callback prop <code>onBookClick()</code> invoked                         | Appointment prompt pops up and confirms slot                    | PASS   |
| Responsive Viewport   | Viewport resized to 375px | CSS Media query <code>@media(max-width: 768px)</code>                    | Grid switches from 3 columns to 1 column stack                  | PASS   |

## 4. Conclusion & Verification

Experiment 03 successfully demonstrated React 18 component-driven architecture. The Virtual DOM reconciled state transitions instantly without whole-page refreshes, props flowed predictably from parents to children, and the responsive design scaled flawlessly across mobile and desktop interfaces.